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Experiment: 8

Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim

To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Objectives

1. To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.
2. To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.
3. To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

Software Required

- **MATLAB**

Theory

MIMO (Multiple-Input Multiple-Output) beamforming enhances wireless communication by employing multiple antenna elements to direct transmitted energy toward a desired user. By adjusting the steering angle through appropriate phase shifts, the beam can be focused in different directions, improving signal strength and reducing interference. Important performance parameters include steering angle, array gain, and beam direction error, which collectively determine the effectiveness of beam steering in modern wireless systems such as 5G NR.

Objective – 1

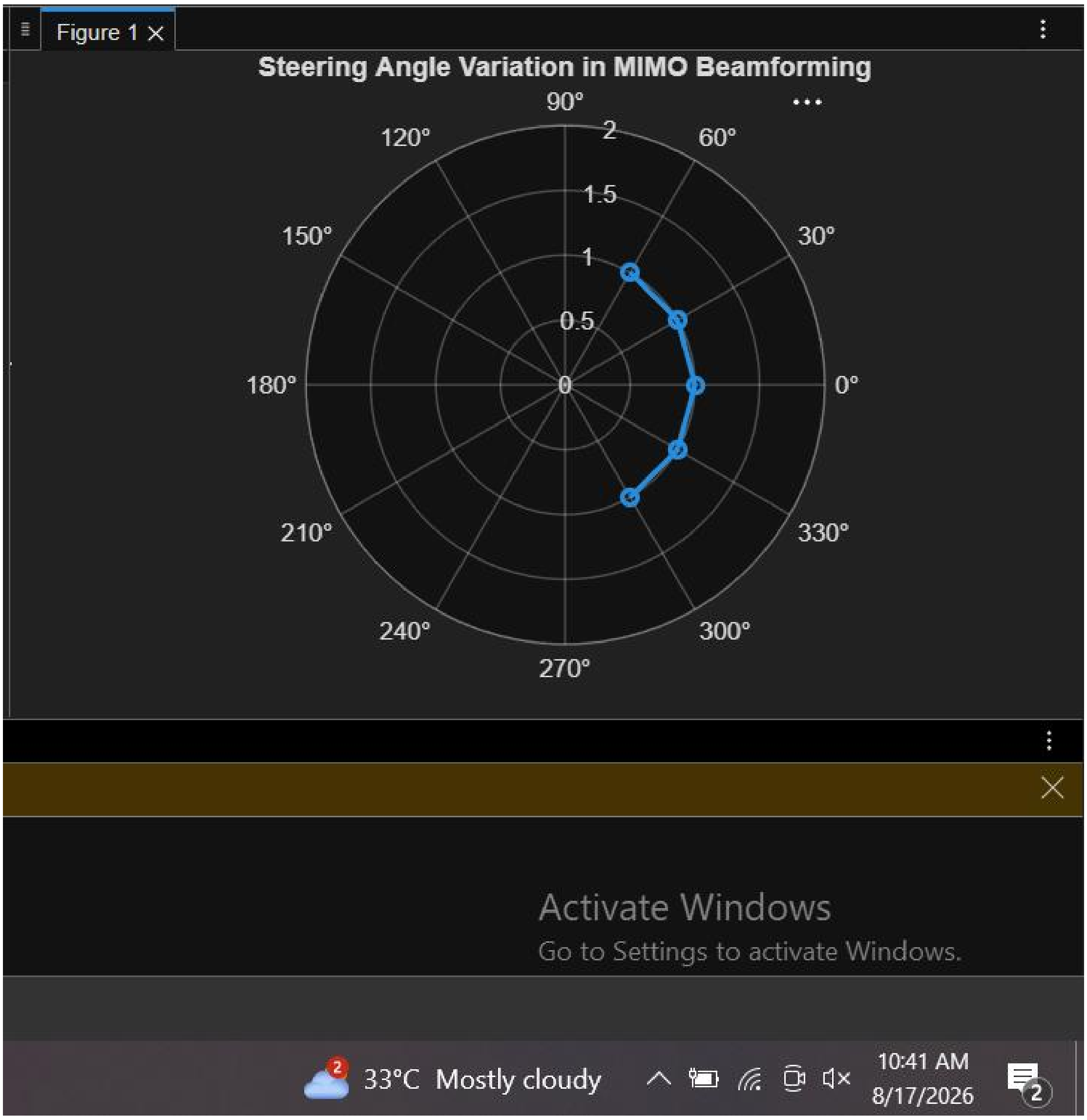
Analyze Steering Angle (Degrees)

Analyze different steering angles of a MIMO beamforming system using MATLAB. Observe how the beam can be directed toward various target directions by varying the steering angle.

MATLAB Code:

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = ones(size(angle)); % Normalized Gain
figure
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)
title('Steering Angle Variation in MIMO Beamforming')
```

Expected Output



• Plot showing different steering angles.

• Beam steering visualization.

Objective – 2

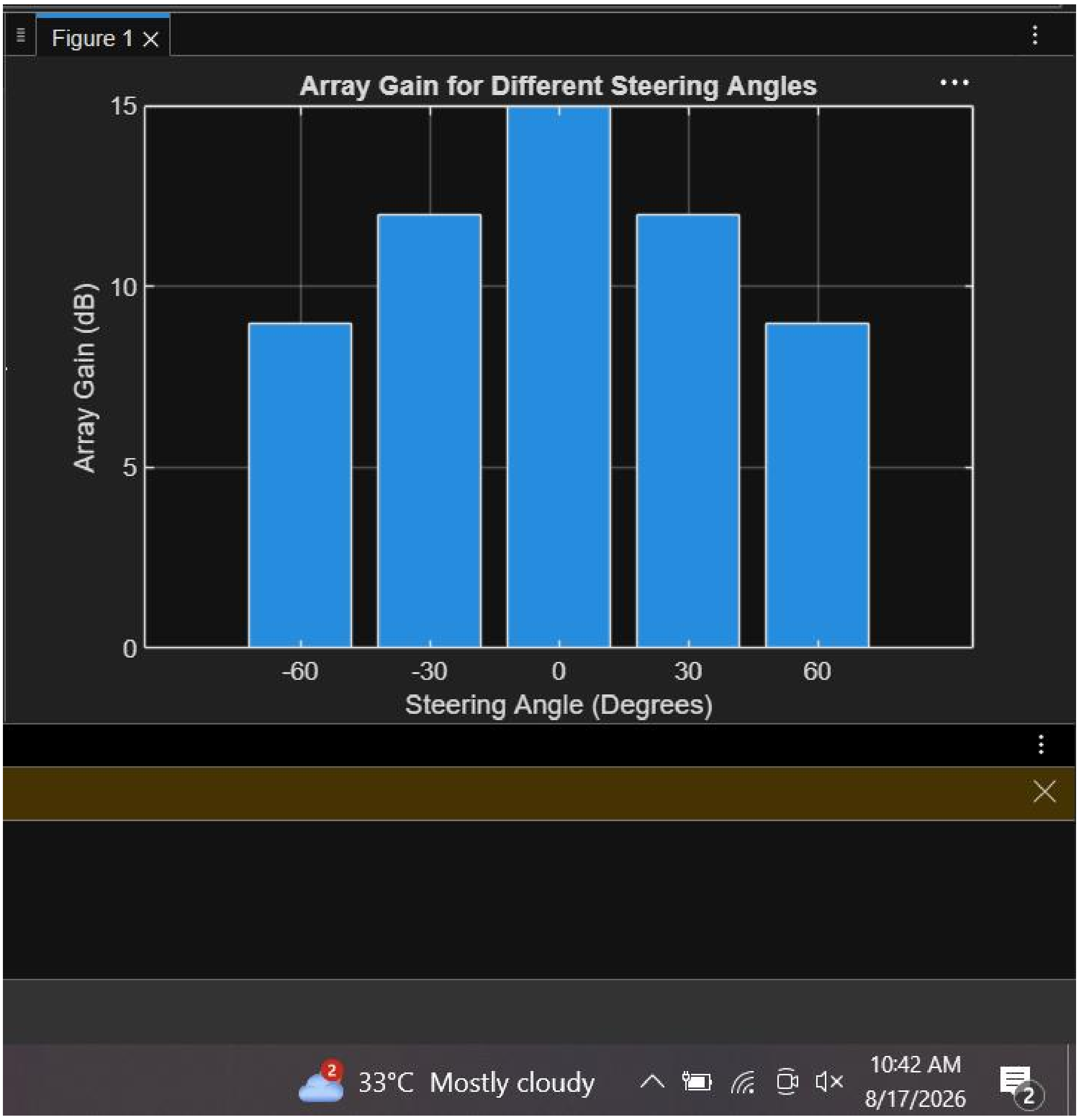
Compare Array Gain (dB)

Compare the array gain obtained at different steering angles using MATLAB. Observe the improvement in signal strength achieved through beamforming.

MATLAB Code:

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = [9 12 15 12 9]; % Array Gain (dB)
figure
bar(angle, gain)
grid on
xlabel('Steering Angle (Degrees)')
ylabel('Array Gain (dB)')
title('Array Gain for Different Steering Angles')
```

Expected Output·



Bar graph showing array gain for different steering angles.

Objective – 3

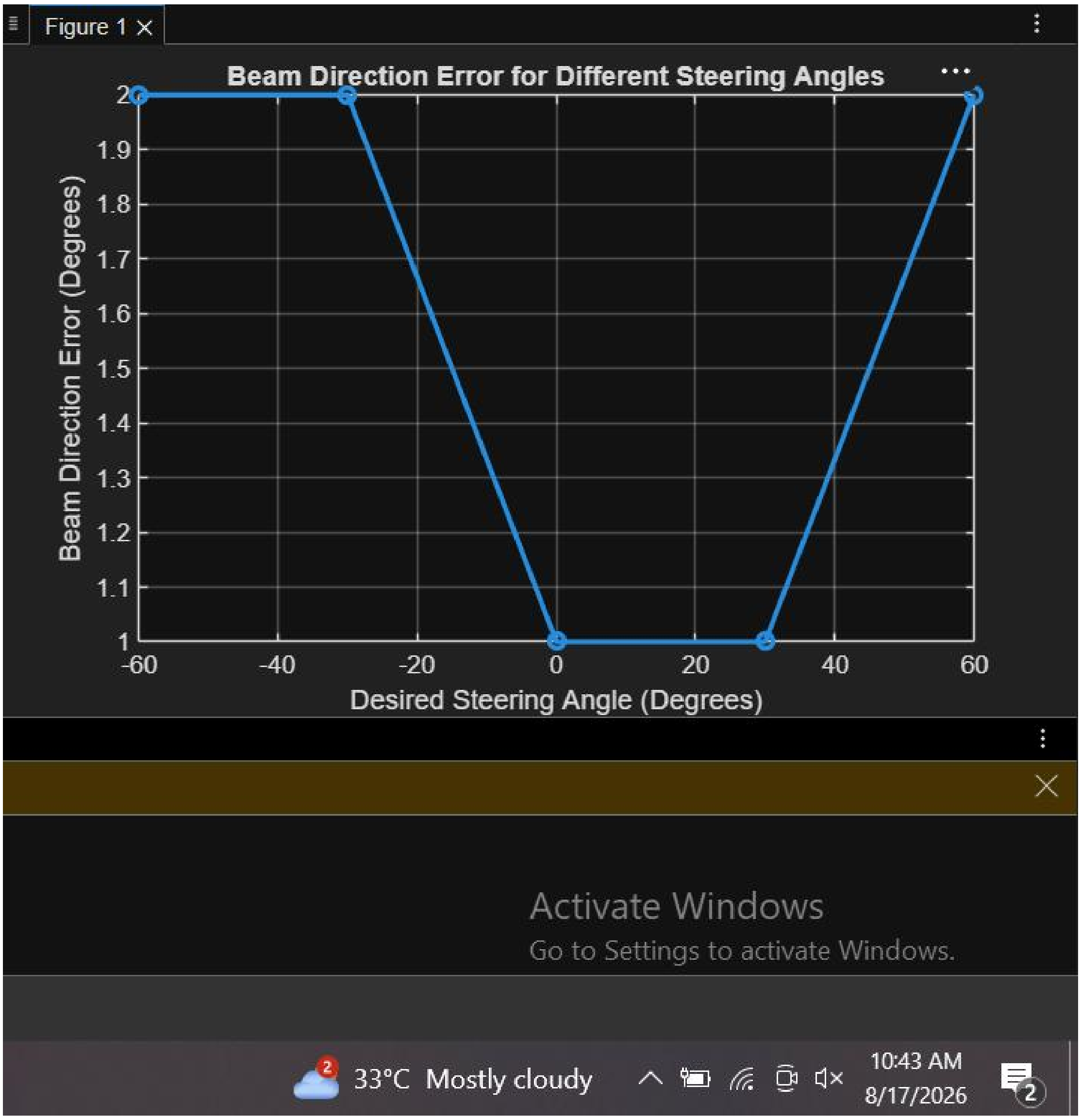
Evaluate Beam Direction Error (Degrees)

Analyze the difference between the desired steering angle and the measured beam direction using MATLAB. Compare the beam direction error for various steering angles and evaluate the beam steering accuracy.

MATLAB Code:

```
clc;
clear;
close all;
desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)
actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)
error = abs(desired - actual); % Beam Direction Error
figure
plot(desired, error, 'o-', 'LineWidth', 2)
grid on
xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')
```

Expected Output·



Line graph showing beam direction error versus steering angle.

Result

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error